# A QUASI-PROJECTIVE MORPHISM OVER A QUASI-COMPACT QUASI-SEPARATED BASE IS AN OPEN IMMERSION INTO A PROJECTIVE MORPHISM

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-quasi-projective-open-projective is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 10717-10722 and proof span 10724-10774 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* Re-checked by hand before carding: this candidate was caught by the lane's thin-proof prefilter because its last sentence contains the phrase 'by definition'. Reading the original shows the proof is a full construction (very ample twist, colimit approximation of the pushforward, lifting of finitely many sections, immersion check, and the quasi-compact immersion step), so the prefilter hit was incidental; the lemma is delivered as the author's own complete argument, and the prefilter reason is recorded here per the fleet rule of 2026-10-14 that a prefilter hit on a proof with at least 800 bytes or 100 lines must be re-read by hand.

*Editorial note.* 8 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- For a finite type morphism over a quasi-compact base a sufficiently high power of an ample sheaf is very ample, cited from the same chapter. Cited at proof lines 10724-10727. Lets the proof replace the given sheaf by a power that is very ample.
- The pushforward of the structure-like sheaf along a quasi-projective morphism is quasi-coherent, and quasi-projective morphisms are quasi-compact and separated, cited from the schemes chapter and the same chapter. Cited at proof lines 10730-10734. Makes the pushforward of the sheaf a quasi-coherent module and gives the quasi-compactness used later.
- A quasi-coherent module is the filtered colimit of its finite type submodules, cited from the properties chapter. Cited at proof lines 10735-10738. Produces the finite type modules over which the construction of the morphism is made.

*Date*: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The description of very ample sheaves on an affine chart by finitely many generating sections and the resulting immersion into a projective space, and permanence of immersions. Cited at proof lines 10744-10749 and 10766-10767. Give the local immersions and show that the globally constructed morphism is an immersion.
- The statement that an immersion which is quasi-compact factors through an open immersion into a closed subscheme of its target, cited from the same chapter. Cited at proof lines 10771-10773. Produces the intermediate scheme X' inside the projective bundle.

*Source reference.* Original lemma 01VU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2459–2459. Same-chapter reference used in the retained author text at source line(s) 9056, 10727. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 07RK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9870–9870. Same-chapter reference used in the retained author text at source line(s) 361, 9510, 9691, 10764. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01RG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2301–2301. Same-chapter reference used in the retained author text at source line(s) 1185, 1204, 2351, 10712, 10770. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01VX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2462–2462. Same-chapter reference used in the retained author text at source line(s) 10733. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 02NP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3488–3488. Same-chapter reference used in the retained author text at source line(s) 9604, 10750. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01PK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2233–2233. Same-chapter reference used in the retained author text at source line(s) 10734. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2081–2081. Same-chapter reference used in the retained author text at source line(s) 262, 404, 882, 1765, 1924, 2256, 2296, 8726, 9175, 10731, 13903. It is supplied only as a navigation pointer so that the retained

text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 03GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4529–4529. Same-chapter reference used in the retained author text at source line(s) 1003, 2371, 4298, 8937, 8952, 9829, 10100, 10769, 13984, 13991, 14143. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $f : X \to S$ be a quasi-projective morphism with $S$ quasi-compact and quasi-separated. Then $f$ factors as $X \to X' \to S$ where $X \to X'$ is an open immersion and $X' \to S$ is projective.*

*Proof.* Let $\mathcal{L}$ be $f$-ample. Since $f$ is of finite type and $S$ is quasi-compact $\mathcal{L}^{\otimes n}$ is $f$-very ample for some $n > 0$, see Lemma 01VU. Replace $\mathcal{L}$ by $\mathcal{L}^{\otimes n}$. Write $\mathcal{F} = f_*\mathcal{L}$. This is a quasi-coherent $\mathcal{O}_S$-module by Schemes, Lemma 01LC (quasi-projective morphisms are quasi-compact and separated, see Lemma 01VX). By Properties, Lemma 01PK we can find a directed set $I$ and a system of finite type quasi-coherent $\mathcal{O}_S$-modules $\mathcal{E}_i$ over $I$ such that $\mathcal{F} = \operatorname{colim} \mathcal{E}_i$. Consider the compositions $\psi_i : f^*\mathcal{E}_i \to f^*\mathcal{F} \to \mathcal{L}$. Choose a finite affine open covering $S = \bigcup_{j=1,\ldots,m} V_j$. For each $j$ we can choose sections

$$s_{j,0}, \ldots, s_{j,n_j} \in \Gamma(f^{-1}(V_j), \mathcal{L}) = f_*\mathcal{L}(V_j) = \mathcal{F}(V_j)$$

which generate $\mathcal{L}$ over $f^{-1}V_j$ and define an immersion

$$f^{-1}V_j \longrightarrow \mathbf{P}^{n_j}_{V_j},$$

see Lemma 02NP. Choose $i$ such that there exist sections $e_{j,t} \in \mathcal{E}_i(V_j)$ mapping to $s_{j,t}$ in $\mathcal{F}$ for all $j = 1, \ldots, m$ and $t = 1, \ldots, n_j$. Then the map $\psi_i$ is surjective as the sections $f^*e_{j,t}$ have the same image as the sections $s_{j,t}$ which generate $\mathcal{L}|_{f^{-1}V_j}$. Whence we obtain a morphism

$$r_{\mathcal{L},\psi_i} : X \longrightarrow \mathbf{P}(\mathcal{E}_i)$$

over $S$ such that over $V_j$ we have a factorization

$$f^{-1}V_j \to \mathbf{P}(\mathcal{E}_i)|_{V_j} \to \mathbf{P}^{n_j}_{V_j}$$

of the immersion given above. It follows that $r_{\mathcal{L},\psi_i}|_{V_j}$ is an immersion, see Lemma 07RK. Since $S = \bigcup V_j$ we conclude that $r_{\mathcal{L},\psi_i}$ is an immersion. Note that $r_{\mathcal{L},\psi_i}$ is quasi-compact as $X \to S$ is quasi-compact and $\mathbf{P}(\mathcal{E}_i) \to S$ is separated (see Schemes, Lemma 03GI). By Lemma 01RG there exists a closed subscheme $X' \subset \mathbf{P}(\mathcal{E}_i)$ such that $i$ factors through an open immersion $X \to X'$. Then $X' \to S$ is projective by definition and we win. □